

M1.3: Impulse and Momentum

1. Momentum and Impulse

Momentum: $\mathbf{p} = m\mathbf{v}$ (kg m/s; vector).

Impulse: $\mathbf{J} = \int \mathbf{F} dt = \Delta\mathbf{p} = m\mathbf{v} - m\mathbf{u}$ (N s).

The impulse of a force equals the change in momentum it produces — a direct consequence of $\mathbf{F} = m\mathbf{a}$ integrated in time.

Conservation of Momentum

For an isolated system (no external forces), total momentum is conserved:

$$\sum m_i \mathbf{u}_i = \sum m_i \mathbf{v}_i.$$

2. One-Dimensional Collisions

Two particles of masses m_1, m_2 travelling along a straight line with velocities u_1, u_2 collide, emerging with velocities v_1, v_2 .

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \text{ (conservation of momentum)}$$

Newton's Experimental Law of Restitution

For a collision between two bodies along the line of centres:

$$\text{speed of separation} = e \times \text{speed of approach}$$

i.e. $v_2 - v_1 = -e(u_2 - u_1)$, where $0 \leq e \leq 1$ is the **coefficient of restitution**.

$e = 1 \rightarrow$ *perfectly elastic* (kinetic energy conserved). $e = 0 \rightarrow$ *perfectly inelastic* (objects stick together).

Kinetic Energy

Total KE = $\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$. In general, KE is lost in collisions unless $e = 1$.

$$\text{KE lost} = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (u_1 - u_2)^2 (1 - e^2)$$

A 2 kg ball moving at 5 m/s collides with a stationary 3 kg ball. Coefficient of restitution $e = 0.4$. Find the velocities after the collision.

Momentum: $2(5) + 3(0) = 2v_1 + 3v_2 \Rightarrow 10 = 2v_1 + 3v_2$.

Restitution: $v_2 - v_1 = 0.4(5 - 0) = 2 \Rightarrow v_2 = v_1 + 2$.

Substitute: $10 = 2v_1 + 3(v_1 + 2) = 5v_1 + 6 \Rightarrow v_1 = 0.8 \text{ m/s}, v_2 = 2.8 \text{ m/s}$.

3. Collisions with a Smooth Wall

Impact is perpendicular to the wall, so the parallel component of velocity is unchanged; the perpendicular component reverses and is multiplied by e :

$$v_{\perp} = -e u_{\perp}, \quad v_{\parallel} = u_{\parallel}.$$

For a ball bouncing vertically with $e < 1$, successive heights form a geometric sequence with ratio e^2 .

A ball dropped from height h bounces to height $e^2 h$.

With e the coefficient of restitution on a flat floor, if dropped from h , it hits with speed $v = \sqrt{2gh}$ and leaves with speed ev , reaching height $h' = (ev)^2 / (2g) = e^2 h$. So successive heights are $h, e^2 h, e^4 h, \dots$

4. Two-Dimensional Collisions (Oblique Impact)

For a ball hitting a smooth wall obliquely, resolve the velocity into components parallel and perpendicular to the wall:

- Parallel: unchanged.
- Perpendicular: magnitude multiplied by e , direction reversed.

For collisions between two spheres, use conservation of momentum *along* the line of centres (with restitution) and *perpendicular* to it (unchanged, assuming smooth spheres).

5. Impulse as a Vector

In 2D/3D problems, impulse $\mathbf{J} = \Delta(m\mathbf{v})$ is a vector. Write components separately.

A ball of mass 0.5 kg has velocity $(4, -3)$ m/s before impact and $(-2, -3)$ m/s after. Find the impulse.

$$\mathbf{J} = 0.5((-2, -3) - (4, -3)) = 0.5(-6, 0) = (-3, 0) \text{ N s.}$$

Magnitude: $|\mathbf{J}| = 3 \text{ N s}$, directed along the negative x -axis.

6. Multi-Body Collisions

With three (or more) bodies, apply conservation of momentum and restitution to each collision in turn, using the post-first-collision velocities as inputs to the next.

7. Exam Style Strategy

- Draw a clear diagram showing velocities before and after, with consistent sign convention.
- Write momentum conservation and Newton's law of restitution as two equations — solve simultaneously.
- For oblique impacts, resolve velocities perpendicular and parallel to the relevant surface/line of centres.
- Check the answer makes physical sense — directions consistent, KE not gained.